

Homework 3

Due Date: 2:00 pm, May 18 (Monday)

Problem 1. For each of the following linear programming models, give your recommendation on which is the more efficient way, probably, to obtain an optimal solution: by applying the simplex method directly to this primal problem or by applying the simplex method directly to the dual problem instead. Explain.

(a) Maximize

$$Z = 8x_1 - 3x_2 + 5x_3,$$

subject to

$$2x_1 - x_2 + 3x_3 \leq 28,$$

$$x_1 + 2x_2 + x_3 \leq 34,$$

$$4x_1 + x_2 + 2x_3 \leq 50,$$

$$3x_1 - 2x_2 + x_3 \leq 22,$$

$$x_1 + 3x_2 + 2x_3 \leq 45,$$

$$5x_1 + 2x_2 + x_3 \leq 60,$$

and

$$x_1 \geq 0, \quad x_2 \geq 0, \quad x_3 \geq 0.$$

(b) Maximize

$$Z = 3x_1 + 6x_2 + 2x_3 + 5x_4 + 4x_5 + 7x_6,$$

subject to

$$x_1 + 2x_2 + 3x_3 + x_4 + 4x_5 + 2x_6 \leq 18,$$

$$3x_1 + x_2 + 2x_3 + 5x_4 + x_5 + 4x_6 \leq 24,$$

and

$$x_j \geq 0, \quad \text{for } j = 1, 2, \dots, 6.$$

Problem 2. Consider the following problem.

$$\text{Maximize } Z = -2x_1 - x_2 - 3x_3,$$

subject to

$$2x_1 + x_2 + x_3 \leq 15,$$

$$x_1 + 3x_2 - 2x_3 \leq 4,$$

and

$$x_1 \geq 0, \quad x_2 \geq 0, \quad x_3 \geq 0.$$

- (a) Construct the dual problem.
- (b) Use duality theory to show that the optimal solution for the primal problem has $Z \leq 0$.

Problem 3. Consider the following problem.

$$\text{Maximize } Z = 4x_1 - 12x_2,$$

subject to

$$x_1 - 3x_2 \leq 2,$$

and

$$x_1 \geq 0, \quad x_2 \geq 0.$$

- (a) Construct the dual problem, and then find its optimal solution by inspection.
- (b) Use the complementary slackness property and the optimal solution for the dual problem to find the optimal solution for the primal problem.
- (c) Suppose that c_1 , the coefficient of x_1 in the primal objective function, actually can have any value in the model. For what values of c_1 does the dual problem have no feasible solutions? For these values, what does duality theory then imply about the primal problem?

Problem 4. Consider the following linear programming problem.

$$\begin{array}{ll}\text{Maximize} & Z = 3x_1 - 2x_2 + x_3, \\ \text{subject to} & x_1 + x_2 - x_3 = 6, \\ & 2x_1 - x_2 + 3x_3 \leq 10, \\ & -x_1 + 4x_2 + x_3 \geq -2, \\ \text{and} & x_1 \text{ is unrestricted in sign,} \\ & x_2 \geq 0, \\ & x_3 \leq 0.\end{array}$$

- (a) Use the dual construction rules directly to construct the dual problem.
- (b) Convert the primal problem to standard maximization form, construct the corresponding dual problem, and show that this dual problem is equivalent to the one obtained in part (a).
- (c) Verify that

$$\bar{x} = \left(\frac{26}{5}, \frac{4}{5}, 0 \right)$$

and

$$\bar{y} = (2, 0, -1)$$

are optimal solutions for the primal and dual problems, respectively.